EXPERIMENT – 1

Study of Computer Forensics and Different Tools Used for Forensic Investigation

Aim

To study the field of computer (digital) forensics and to understand the different tools that are used during a forensic investigation.

Introduction

Digital forensics is the field of determining who was responsible for a digital intrusion or other computer crime. It uses a wide range of techniques to gain attribution to the perpetrator. It relies on the fundamental concept that whenever a digital intrusion or a crime is committed, the perpetrator inadvertently leaves a bit of themselves behind for the investigator to find. These 'bits' may be entries in log files, changes to the registry, hacking software, malware or remnants of deleted files, and all of them can provide clues and evidence that help determine the identity of the offender.

Just like hacking, there are a number of software tools for performing digital forensics. Most investigators rely on three major commercial suites – EnCase Forensic (Guidance Software), Forensic Tool Kit or FTK (Access Data) and ProDiscover. These suites are expensive and are mainly used by law enforcement. The same or similar techniques are available in free and open-source suites such as The Sleuth Kit (TSK), Helix and Knoppix, without the polished interface. In addition, a large number of individual tools are available in distributions such as BackTrack and Kali Linux.

Procedure

- The commercial forensic suites used by law enforcement – EnCase Forensic, Forensic Tool Kit (FTK) and ProDiscover – were identified.
- The free and open-source forensic suites were studied – The Sleuth Kit (TSK), Helix and Knoppix.
- The individual forensic tools available in the Kali / BackTrack distribution were listed and their purpose was noted.

- The forensics field tools such as FTK Imager, Log Parser Lizard GUI, Noxcivis Field Toolkit, Active@ Partition Recovery and Autopsy were studied.

The function of each tool identified during the study is summarised in the table below.

Forensic Tools Studied

Tool	Purpose
EnCase Forensic	Commercial forensic suite widely used by law enforcement.
FTK (Forensic Toolkit)	Commercial suite; FTK Imager creates forensically sound disk images.
ProDiscover	Commercial forensic investigation suite.
The Sleuth Kit (TSK)	Open-source collection of command-line forensic tools.
Autopsy	Graphical front-end for The Sleuth Kit; used to investigate disks and images.
FTK Imager	Forensic disk imager, memory capture and file recovery.
Log Parser Lizard GUI	Flexible and powerful log file parser.
Noxcivis Field Toolkit	Collects information from a live computer during incident response.
Active@ Partition Recovery	Recovers deleted and lost partitions.
USBDeview	Lists connected and previously used USB devices.
LastActivityView	Displays a log of recent user actions and system events.
ChromeHistoryView / IECacheView	Reads browser history and cache artifacts.
Compare It	Compares two files and highlights the differences.
DumpIt	Generates a physical memory dump of Windows machines.
Bulk Extractor	Extracts features such as e-mail, URLs and phone numbers from an image.
ExifReader	Extracts EXIF metadata from image files.
Recover My Files	Recovers deleted files from drives and disk images.

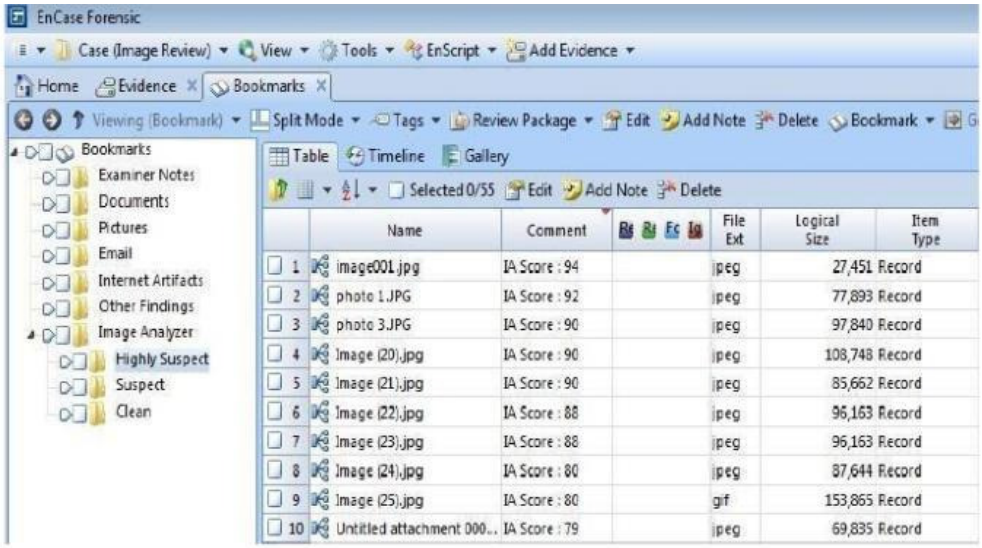


Fig. 1.1 EnCase Forensic – a widely used commercial forensic suite
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- sleuthkit
- truecrypt
- hexedit
- autopsy
- iphoneanalyzer
- rifiuti2
- ptk
- exiftool
- evtparse.pl
- fatback
- scalpel
- dc3dd
- driftnet
- timestomp

Fig. 1.2 Some of the forensic tools available in the BackTrack / Kali distribution
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Result

Thus, the field of computer forensics and the various tools used for forensic investigation were studied successfully, along with their purpose and usage.